

# BUSINESS ANALYST PROJECT CANVAS

## Library Management System

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| <b>CONCEPT</b><br><i>Describe the idea of the web app you want to create.</i><br><br>A browser-based library management system that runs the whole lending cycle in one place: a public catalogue members search from any device, a staff console for issue, return, cataloguing and stocktake, and a management view for acquisitions and reporting.<br><br>It replaces paper registers and spreadsheets, and works on the desk PC, a self-service kiosk and a phone without installing anything.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>OBJECTIVES &amp; PURPOSES</b><br><i>What goal do you want to achieve? It must be measurable and shared with the team.</i> <ul style="list-style-type: none"><li>• One searchable catalogue covering physical and digital holdings.</li><li>• 95% of loans processed through self-service within 6 months.</li><li>• Average issue time under 30 seconds; catalogue search under 2 seconds.</li><li>• Overdue returns cut by 30% through automated reminders.</li><li>• 90% of holdings barcoded and reconciled at the first stocktake.</li></ul>                                                                                                                                                                                                                                                                                                                                 | <b>VALUES</b><br><i>What creates real value for your user and ties them to your product? Max 3 themes.</i> <ul style="list-style-type: none"><li>• Find it fast — one search, live availability, exact shelf location.</li><li>• Borrow without asking — self-issue, renewal and holds from any browser.</li><li>• No surprise fines — reminders before the due date, not after.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>PERSONAS</b><br><i>Describe the archetype of the users who will use your web app.</i><br><br><b>Students</b> <ul style="list-style-type: none"><li>— Borrow textbooks, queue for high-demand reserve items each semester.</li></ul><br><b>Academic staff &amp; researchers</b> <ul style="list-style-type: none"><li>— Long loans, reading lists, inter-library requests.</li></ul><br><b>Circulation &amp; cataloguing staff</b> <ul style="list-style-type: none"><li>— Issue, return, shelve, catalogue, chase overdues. Heaviest daily users.</li></ul><br><b>Library manager</b> <ul style="list-style-type: none"><li>— Acquisitions, budget, membership and usage reporting.</li></ul><br><b>System administrator</b> <ul style="list-style-type: none"><li>— Accounts, roles, integrations, backups.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>PROBLEMS TO SOLVE</b><br><i>Weaknesses of the current / similar solution. What is not working?</i> <ul style="list-style-type: none"><li>• Loan records kept in registers and spreadsheets are inconsistent and impossible to audit.</li><li>• Members cannot check availability without walking to the shelf.</li><li>• Overdue tracking and fine calculation are manual, so they get skipped.</li><li>• Reservations are handled informally; popular titles go to whoever asks loudest.</li><li>• Stocktake takes weeks and still misses lost copies.</li><li>• No usage data to justify purchases or weed dead stock.</li><li>• Staff-only desktop tools mean nothing is reachable off campus.</li></ul> | <b>COMPONENTS</b><br><i>What are the main modules that make up the app? Put all features into a defined component.</i><br><br>Catalogue & Cataloguing · Circulation · Membership · Reservations · Fines & Payments · Acquisitions · Reporting · Administration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>FEATURES</b><br><i>What should the app do? What services are provided? List what users expect to find.</i><br><br><b>Catalogue</b> <ul style="list-style-type: none"><li>— Search by title, author, ISBN or subject; filters for format and branch; live availability and shelf location; MARC record import; copy-level accession records.</li></ul><br><b>Circulation</b> <ul style="list-style-type: none"><li>— Barcode / RFID issue, return and renewal; kiosk self-service mode; loan periods and limits by borrower category; offline desk mode that syncs when the network returns.</li></ul><br><b>Reservations</b> <ul style="list-style-type: none"><li>— Hold queue for on-loan titles, hold-shelf notification, expiry of uncollected holds.</li></ul><br><b>Members</b> <ul style="list-style-type: none"><li>— Registration, card issue, borrowing history, suspension rules, self-service profile.</li></ul><br><b>Fines &amp; payments</b> <ul style="list-style-type: none"><li>— Automatic overdue calculation, waiver workflow with reason codes, online card payment and receipts.</li></ul><br><b>Acquisitions</b> <ul style="list-style-type: none"><li>— Purchase requests from staff and members, order tracking, budget spend against fund lines.</li></ul><br><b>Reporting</b> <ul style="list-style-type: none"><li>— Circulation volume, popular and dead stock, overdue ageing, membership growth, stocktake discrepancy report; CSV export.</li></ul><br><b>Notifications</b> <ul style="list-style-type: none"><li>— Email and SMS for due dates, overdues, holds ready and account changes.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>MILESTONES</b><br><i>Define the roadmap of the project. Start from the MVP and then scale.</i><br><br><b>Month 1 — Discovery</b> <ul style="list-style-type: none"><li>— Requirements elicitation, as-is process mapping, MVP scope and wireframes signed off.</li></ul><br><b>Month 2 — MVP core</b> <ul style="list-style-type: none"><li>— Catalogue and membership modules; legacy data cleansed and trial-migrated.</li></ul><br><b>Month 3 — Lending</b> <ul style="list-style-type: none"><li>— Circulation, reservations and fines; barcode / RFID hardware integrated.</li></ul><br><b>Month 4 — Scale</b> <ul style="list-style-type: none"><li>— Acquisitions, reporting, kiosk build; staff training and UAT.</li></ul><br><b>Month 5 — Launch</b> <ul style="list-style-type: none"><li>— Pilot branch, full migration, cut-over, go-live with hypercare.</li></ul> | <b>DELIVERABLES</b><br><i>What will the team have to produce, besides the app itself?</i><br><br><b>Analysis</b> <ul style="list-style-type: none"><li>— User stories (INVEST, Given-When-Then)</li><li>— Requirement traceability matrix</li><li>— As-is / to-be process flows</li><li>— ER model &amp; data dictionary</li><li>— Business rules catalogue</li></ul><br><b>Design &amp; build</b> <ul style="list-style-type: none"><li>— Wireframes and clickable prototype</li><li>— API documentation</li><li>— Data migration &amp; cleansing plan</li></ul><br><b>Quality &amp; handover</b> <ul style="list-style-type: none"><li>— Test cases, defect log, UAT sign-off</li><li>— Staff training pack &amp; user manual</li><li>— Support and backup runbook</li></ul> |

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| <div>STAKEHOLDERS</div> <div>Who needs to be involved? What is their role and who makes the decisions?</div> <div>Internal<ul style="list-style-type: none"><li>Library manager — sponsor, owns loan policy and sign-off.</li><li>Circulation &amp; cataloguing staff — daily users, define desk workflow.</li><li>IT &amp; Security — SSO, hosting, privacy gatekeepers.</li><li>Finance — fine income, acquisition budget.</li><li>Delivery team — BA, PM, developers, UI/UX, QA.</li></ul></div> <div>External<ul style="list-style-type: none"><li>Members — students, staff, public borrowers.</li><li>Book vendors &amp; e-resource publishers.</li><li>Inter-library loan network partners.</li><li>Payment gateway provider.</li><li>Hardware suppliers — scanners, kiosks, security gates.</li></ul></div> |  | <div>RISKS</div> <div>What could go wrong and stop the goal being achieved?</div> <ul style="list-style-type: none"><li>Data migration — legacy records are duplicated and non-standard; cleansing must finish before cut-over.</li><li>Barcode / RFID gaps — unlabelled or damaged tags stall the desk.</li><li>Privacy — borrowing history is sensitive personal data under the Australian Privacy Principles.</li><li>Staff adoption — desk staff fall back to paper if training is thin.</li><li>Availability — a browser-only system stops lending during an outage; offline mode is mandatory.</li><li>Fine disputes — wrong calculations create refunds and lost trust.</li><li>Browser fragmentation — old machines on the public floor may run outdated browsers.</li></ul> |  |
| <div>CONTEXT OF USAGE &amp; COVERAGE</div> <div>Where will the app be used? Always online, or offline mode too?</div> <ul style="list-style-type: none"><li>Circulation desk and self-service kiosks during opening hours.</li><li>Handheld scanners in the stacks for shelving and stocktake.</li><li>Members search and renew from home, campus Wi-Fi and mobile data.</li><li>Offline mode at the desk, queued and synced on reconnect.</li><li>Peak load at semester start, end and assignment deadlines.</li></ul>                                                                                                                                                                                                                                                                                             |  | <div>TECHNOLOGY</div> <div>Stack, frameworks and integrations.</div> <ul style="list-style-type: none"><li>Frontend: React (SPA), responsive layout.</li><li>Backend: Node.js / Express REST API.</li><li>Database: PostgreSQL with full-text search indexing.</li><li>Auth: university SSO with role-based access.</li><li>Integrations: payment gateway, email / SMS, MARC import, scanner and receipt-printer drivers.</li></ul>                                                                                                                                                                                                                                                                                                                                                  |  |
| <div>RESPONSIVENESS &amp; ACCESSIBILITY</div> <div>Layout behaviour and standards to meet.</div> <ul style="list-style-type: none"><li>Mobile-first for the member catalogue; fixed wide layout for the staff console.</li><li>Kiosk view: large touch targets, no keyboard needed.</li><li>WCAG 2.1 AA — keyboard navigation, contrast, screen-reader labels.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                            |  | <div>BROWSERS &amp; DEVICES</div> <div>Which browsers, devices and screen sizes must be supported?</div> <ul style="list-style-type: none"><li>Chrome, Edge, Firefox and Safari — current and one previous version.</li><li>Staff desktops (Windows), kiosk terminals in full-screen mode.</li><li>Android handheld scanners for stocktake.</li><li>Member access on phone, tablet and laptop.</li></ul>                                                                                                                                                                                                                                                                                                                                                                             |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | <div>HOSTING &amp; RELEASE</div> <div>Where does it live and how is it shipped?</div> <ul style="list-style-type: none"><li>Hosted on university-managed cloud infrastructure inside Australia.</li><li>Public catalogue at a public URL; staff console behind SSO on an internal route.</li><li>CI/CD pipeline with staging and production; fortnightly releases.</li><li>Phased rollout: pilot branch first, then all branches; legacy register kept read-only for one semester.</li></ul>                                                                                                                                                                                                                                                                                         |  |